

1.0 Background

Nigeria has over 184 million active telecom subscribers who rely on telecommunications services for calls, internet access, banking, education, healthcare, e-commerce, and business activities. Every day, telecom providers receive thousands of customer complaints through customer care services, emails, social media, mobile applications, and websites.

Although many organizations have adopted digital customer service platforms, complaint handling often still depends on manual review, predefined routing rules, keyword-based classification, and human judgment. These approaches can struggle with large complaint volumes and with interpreting language dialects, slangs, misspelling, and context, which has led to delays and misclassification of customer complaints. This has motivated the development of AI based complaint classification systems that better understand natural language.

These methods struggle to understand how Nigerians naturally communicate. For example, a customer may write: "My line no dey work."
Another may write: "Airtime wey i put now now don finish pata pata."
Another may write: "MTN una don swallow my data."

Although all three complaints describe shows Nigerian expressions in the complaints, a keyword-based system may classify them differently or fail to recognize the three of them entirely.

2.0 Problem Statement

The problem this project is trying to solve is that existing complaint management systems cannot accurately understand the diversity of Nigerian customer language, resulting in inefficient complaint classification, delayed service delivery, increased operational costs, and poor customer satisfaction.

3.0 The Approach

The project focuses on natural language understanding rather than simple keyword matching.

Instead of asking: "Does this sentence contain the word 'network'?"
the system asks: "What is the customer actually complaining about?"

The approach involves:

Step 1: Collect real customer complaints and datasets. From

- Social media accounts
- Customer care logs
- Complaint portals
- Telecom forums

Step 2: Prepare the data. This includes:

- Cleaning of the datasets.
- Removing duplicates.
- Correcting obvious errors.
- Protecting user privacy

Step 3: Understanding Nigerian language patterns. The system learns:

- Nigerian English
- Pidgin
- Abbreviations
- Misspellings
- Language mixing (multilingual expression)

Step 4: Assigning of complaints to categories. List of the categories are:

- Network failure
- Billing
- Data issues
- SIM registration
- Call quality
- Airtime deduction
- Mobile money
- Customer service

Step 5: Send each complaint to the appropriate department.

4.0 Proposed Solution

The proposed solution is an AI-assisted complaint classification system designed specifically for Nigerian telecommunications.

Instead of relying only on keywords, the system understands the meaning and the context of customers complaint, even when written in Nigerian English, Pidgin, abbreviations, or with misspelling. It automatically classifies the complaints into their appropriate category, allowing telecom providers to direct them to the correct support teams more quickly.

Key Features

- Ability to understand Nigerian communication styles.
- Automatically classifies complaints.
- Directs complaint to the right department.
- Reduces manual workload.
- Improves response times.
- Maintains privacy by protecting customer information.
- Supports human review when confidence is low.

5.0 Ideology

The ideology describes the principles that shape how the solution is developed and used.

A. Fairness

Every customer deserves equal service regardless of: their language, background, knowledge or dialect. A complaint written in Pidgin should receive the same attention as one written in English also a complaint written in Yoruba should receive the same attention as one written in Hausa.

B. Accessibility

Technology should work for everyone. The system should recognize Nigerian English, Pidgin, local expressions, common abbreviations, indigenious language.

C. Privacy

Customer information must be protected. Personal details should be anonymized where possible, and complaint data should be handled securely.

D. Transparency

The system should support, not replace, human decision-making. Customer service agents should understand why a complaint was classified in a certain way, and there should be a process for reviewing uncertain cases.

E. Accountability

Humans remain responsible for final decisions, especially for sensitive or ambiguous complaints.

F. Reliability

The system should consistently perform well across different complaint categories and language styles.

5.1 Expected Impact

1. **For Customers:** The system should be able to provide:

- Faster responses.
- Shorter waiting times.
- Better customer experience.
- Increased trust in telecom services.

2. **For Telecom Companies:** The system should be able to provide:

- Reduced manual workload.
- Lower operational costs.
- Improved complaint management.
- Better service quality.

3. **For Regulators**

Organizations like the Nigerian Communications Commission can more easily identify recurring issues and monitor service quality trends.

4. **For Society**

- Improved digital service delivery.
- Greater inclusion for users with diverse language backgrounds.
- More equitable access to customer support.
- Stronger confidence in AI-assisted public-facing services.

6.0 Summary

This project aims to improve customer support in Nigeria's telecommunication sector by accurately understanding and classifying complaints written in diverse local languages, enabling faster resolution, fair services, lower operational costs, and a more inclusive customer experience.

References

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